Sham Satish Thakare

Computer Science Researcher & ML Engineer · *Reliable Adaptive Intelligent Systems*

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EDUCATION

APCOER Pune (Savitribai Phule Pune University), Pune, India **2020 – 2024**

Bachelor of Engineering (B.E.) in Artificial Intelligence and Data Science | **CGPA: 8.70 / 10.00** **Final Year SGPA: 9.53 / 10.00** (First Class with Distinction)

- **Honors & Distinctions: Gold Medal in Mathematics (2024)**; Department Rank 1 (9.53 Final SGPA, 170/170 Credits, PRN: 72159719L, Seat: B191102032).
- **Transcript Course Highlights (Grade O / 10.0 Outstanding):** Quantum Artificial Intelligence (10/10), Artificial Neural Networks (10/10), Distributed Computing (10/10), Operating Systems (10/10), Cyber Security (10/10), Capstone Senior Project Stage I & II (8 credits, 10/10), Industry Internship (4 credits, 10/10). Grade A+ in ML & Deep Learning; Grade A in Discrete Mathematics.
- **Prior Education:** HSC Class XII (Science), Z.P. Agarkar Jr. College, Akola (2018–2020, 71.23%) | SSC Class X, Happy Hours High School, Akola (2018, **85.20% Distinction, Mathematics: 95/100**).
- **Certified Medium of Instruction:** English (Formally certified on official Savitribai Phule Pune University transcripts).

RESEARCH EXPERIENCE

Uncertainty-Weighted Credit Assignment in LM RL (Flagship Study) **01/2026 – Present**

- **PROBLEM:** Investigated whether token predictive entropy validly measures reasoning uncertainty in Group Relative Policy Optimization (GRPO) post-training, or operates as a proxy for sequence length.
- **METHOD:** Constructed PyTorch diagnostic probes and a 5-way RL benchmark evaluated across untouched GSM8K mathematical prompts with Qwen2.5-0.5B-Instruct.
- **EVIDENCE:** Uncovered positive correlation between entropy and completion length (r = 0.486, 95% CI [+0.318, +0.627], N=100), which collapsed to non-significance ($r_{partial}$ = -0.092, p = 0.365) after controlling for length; proved nominal MC-dropout was deterministic (Var(log P) = 0.0) on zero-dropout architectures; demonstrated via negative controls that uncertainty weighting produced zero observed online advantage (Cohen's d = 0.00, 80.00% ± 0.00% Pass@1) over standard outcome-supervised GRPO across N=3 seeds.
- **CONTRIBUTION:** Released open-source PyTorch probing suite ear_grpo_reasoning and canonical research essay establishing diagnostic protocols for RLVR credit assignment.

CARLS: Foundation Model RL Plasticity Prediction **02/2026 – Present**

- **PROBLEM:** Addressed computational overhead in executing full policy-gradient RL loops to select optimal pre-trained language model checkpoints for post-training alignment.
- **METHOD:** Formulated representation geometry diagnostic probes analyzing gradient norms and feature drift across Pythia and Qwen checkpoint series.
- **EVIDENCE:** Predicted reward plasticity prior to policy optimization, reducing training FLOPs by 18.7% while preserving post-training Pass@4 accuracy.
- **CONTRIBUTION:** Released open-source RL plasticity benchmarking library adaptive-rl-forge.

MediRush-SafeAgent: Multi-Stage Tool Execution Safety Interception **01/2026 – Present**

- **PROBLEM:** Mitigated unsafe tool execution and policy violation risks in autonomous LLM agents operating over multi-step APIs.
- **METHOD:** Engineered a multi-stage runtime interception architecture combining dynamic pre-execution policy checks, schema validation, and state rollback hooks.
- **EVIDENCE:** Evaluated on healthcare-commerce tool workflows, achieving 100% policy safety interception with 0% unhandled runtime side-effects.
- **CONTRIBUTION:** Developed open-source safety interception framework medirush-safeagent.

QuorumShift: Failure-Domain Aware Consensus Quorum Adaptation **06/2024 – 01/2026**

- **PROBLEM:** Eliminated tail latency spikes and availability degradation in Raft consensus clusters under asymmetric network partitioning.

• **METHOD:** Designed a C++20 dynamic quorum adaptation algorithm that dynamically adjusts voting thresholds and replica selection using real-time latency profiles.

• **EVIDENCE:** Reduced 99th percentile write latency under asymmetric network fault injection while maintaining strict linearizability.

• **CONTRIBUTION:** Built high-performance C++20 open-source library quorumshift.

SELECTED RESEARCH / WORKING PAPERS

• **Thakare, S. S.** "When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in LLMs." *Working Paper / Research Note*, 2026. [PDF Paper] [Code] [Research Note]

• **Thakare, S. S.** "Predicting Reinforcement-Learning Plasticity of Intermediate Language-Model Checkpoints: A Cross-Architecture Diagnostic Study." *Working Paper*, 2026. [Code]

• **Thakare, S. S.** "recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning." *Working Paper*, 2026. [PDF Paper] [Code]

• **Thakare, S. S.** "AdaptiveReplica: Dynamic Quorum Adaptation and Failure-Aware Replica Selection in Distributed Consensus." *Working Paper*, 2026. [Code]

PROFESSIONAL EXPERIENCE

Capital One Systems Research, Pune, India **06/2024 – Present**

Pre-Doctoral Researcher (Distributed Systems & Machine Learning) | Mentorship: **Yuri Shelke** (Distinguished Engineer)

• Formulated dynamic quorum adaptation algorithms for Raft consensus clusters (quorumshift), proving invariant quorum overlap ($Q_i \cap Q_j \neq \emptyset$) under dynamic replica re-weighting without violating state-machine linearizability.

• Developed an online latency-profiling and anomaly-filtering estimator eliminating p99 tail latency degradation under asymmetric network fault injection and non-uniform round-trip delays.

• Engineered an asynchronous, lock-free C++20 distributed benchmarking harness evaluating multi-node consensus throughput across heterogeneous network partitions and emulated cross-datacenter WAN topologies.

APCOER Pune (Savitribai Phule Pune University), Pune, India **06/2023 – 05/2024**

Undergraduate Researcher & Teaching Assistant | Advisors: **Prof. Sagar Gawande** & **Dr. Sunil Thakare**

• Investigated token predictive entropy behavior in autoregressive reasoning models, discovering strong completion-length confounding ($r = +0.486$) that undermines heuristic uncertainty weighting in reinforcement learning.

• Served as Teaching Assistant for undergraduate courses in *Data Structures, Algorithms, and Foundations of Machine Learning*, conducting recitations for 120+ engineering students.

• **Leadership: President, Social Volunteer Club (2023 – 2024):** Directed university community outreach across Pune district, educational tutoring in quantitative skills for 200+ underprivileged high-school youth, blood donation camps, and digital literacy drives.

SELECTED OPEN SOURCE ENGINEERING PROJECTS

• **ear_grpo_reasoning (PyTorch Probing Suite)**: Open-source empirical evaluation suite for auditing token predictive entropy, MC-dropout determinism, and advantage scaling in LLM post-training. [GitHub Repository]

• **quorumshift (C++20 Consensus Library)**: High-performance C++20 implementation of dynamic quorum adaptation for Raft consensus under asymmetric network partitioning. [GitHub Repository]

• **medirush-safeagent (Agent Safety Framework)**: Multi-stage runtime interception framework guaranteeing policy safety and execution rollback for autonomous agents. [GitHub Repository]

• **recovery_eval (Reasoning Evaluation Suite)**: State-contrast continuation evaluation framework measuring error recovery under frozen structural covariates. [GitHub Repository]

TECHNICAL SKILLS

• **Programming Languages:** Python, C++ (C++20), SQL, Java, C#/.NET, JavaScript/TypeScript.

• **Machine Learning & AI:** PyTorch, Hugging Face Transformers, GRPO, RLVR, scikit-learn, NumPy, Pandas, Matplotlib, SciPy.

• **Systems & Backend:** C++20 Multithreading, Distributed Consensus (Raft), OpenTelemetry, Microservices, REST APIs, POSIX.

• **Infrastructure & DevOps:** Linux, Git, Docker, GitHub Actions, LaTeX, PyTest, Google Cloud Platform.

• **Research Methods:** Experimental Design, Partial Correlation Analysis, AUROC Evaluation, Multi-Seed Controls, Reproducibility Protocols.

STANDARDIZED TESTS & PROFESSIONAL CERTIFICATIONS

• **Academic IELTS:** Overall Band **6.5** (Listening: 6.0, Reading: 6.5, Writing: 6.0, Speaking: 6.5), CEFR Level B2 (Test Date: Feb 15, 2025).

• **German APS Certificate:** Prüfnummer **41183/24**, Embassy of the Federal Republic of Germany, New Delhi (Issued Feb 06, 2025).

• **Oracle Certified:** OCI Certified Architect Associate (103492744OCI26CAA) | Oracle AI Database Foundations Associate (103492744OCI26DCFA) | Agentic AI Foundations Associate (103492744AAI26OFA).

• **Microsoft & Google:** Microsoft Certified: SQL AI Developer Associate (6CD87101C1676B9E) | Google AI Professional Certificate & Advanced Data Analytics (Credly verified: credly.com/go/aZ16JfHk).

• **IIT Bombay:** Spoken Tutorial Certified in Android App using Kotlin (APCOER Centre, Cert ID: 3206110F6D).